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Introduction to Scope Measurements		

Report History

Report 1		Date	Remarks
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Final decision:

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1 Introduction

This lab focused on understanding the working principles and practical applications of semiconductor diodes. Two types of diodes were examined: a germanium diode (AA138) and a silicon diode (1N4148). These components are fundamental in electronics due to their ability to allow current to flow in only one direction.

In the first task, we studied the forward-biased behavior of both diodes. Using an oscilloscope in X-Y mode, we plotted the current-voltage (I-V) characteristics. This allowed us to clearly observe the differences between the germanium and silicon diodes, particularly the forward voltage threshold required for conduction.

In the second task, we applied a DC voltage and plotted the I-V data on semilogarithmic paper. This method enabled us to test the validity of the Shockley diode equation and determine key parameters such as the ideality factor, saturation current, and series resistance. These values provided deeper insight into the electrical behavior of real diodes in circuits.

The third task involved constructing a half-wave rectifier circuit using the silicon diode (1N4148). We measured the output voltage both with and without a smoothing capacitor. From this, we calculated the ripple voltage, which indicated how much the output voltage fluctuated. The presence of the capacitor significantly reduced the ripple, demonstrating its effectiveness in stabilizing the DC output.

In the fourth and final task, we assembled a full-wave rectifier using two diodes. This configuration is more efficient and delivers a more stable DC output. We tested the circuit with different capacitor values and compared the resulting ripple voltages. The results showed that the full-wave rectifier produced less ripple than the half-wave rectifier, highlighting its superior performance in power supply applications.

These four tasks effectively bridged theoretical knowledge with practical experimentation. They enabled us to measure diode behavior, evaluate diode parameters, and implement diodes in useful rectifier circuits.

2 Objective

The main goals of this lab:

- *To observe and compare how a germanium diode (AA138) and a silicon diode (1N4148) behave when forward biased, by measuring their current-voltage (I-V) curves using an oscilloscope in X-Y mode.

- *To analyze the DC behavior of the diodes by plotting their I-V data on semilog paper and calculating key values like the ideality factor, saturation current, and series resistance.

- *To build a half-wave rectifier using the silicon diode, check how the output voltage looks with and without a smoothing capacitor, and calculate the ripple voltage to see how filtering improves the output.

To create a full-wave rectifier with two diodes and compare its performance to the half-wave version by looking at the waveforms and ripple for different capacitor values.

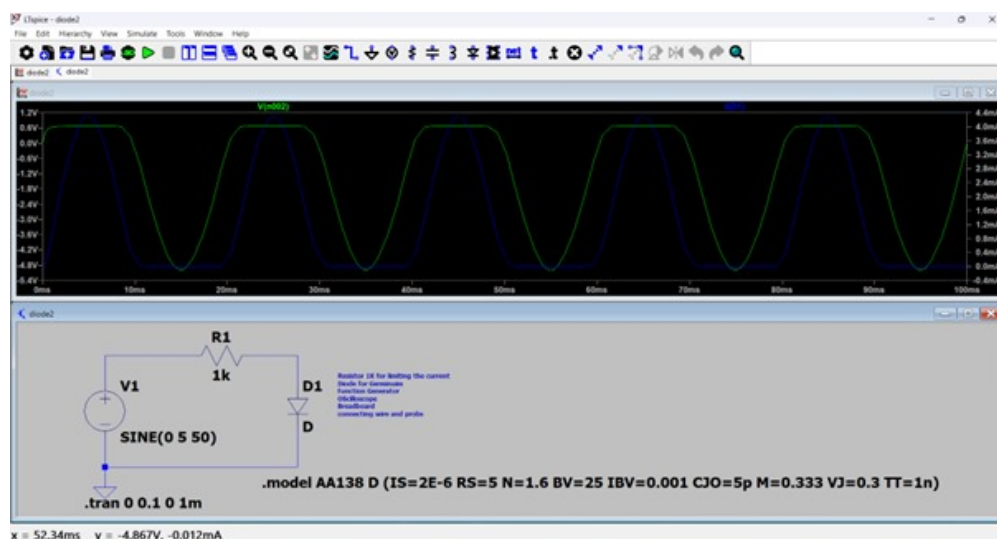
3 Tools and Measurement Equipment

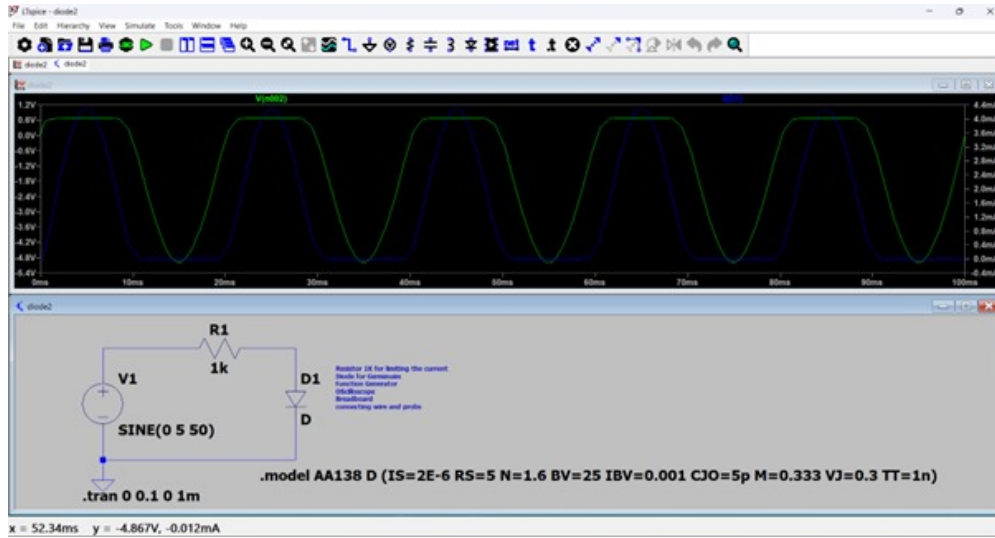
The following tools, equipment, and components were used during the laboratory session for all four tasks:

- Breadboard
- Connecting wires and probes
- Power supply
- Function generator (used for signal input)
- Oscilloscope (dual channel, for waveform observation)
- Digital multimeter (for voltage and current measurements)
- Resistors (100 Ω , 1 k Ω , 47 Ω)
- Diodes:
 - Germanium diode (AA138)
 - Silicon diode (1N4148)
- Electrolytic capacitors (100 μ F and 470 μ F)
- LTSpice simulation

Note: Placeholders are included here for additional graphs and analysis plots.

4 Task 1 Preparation



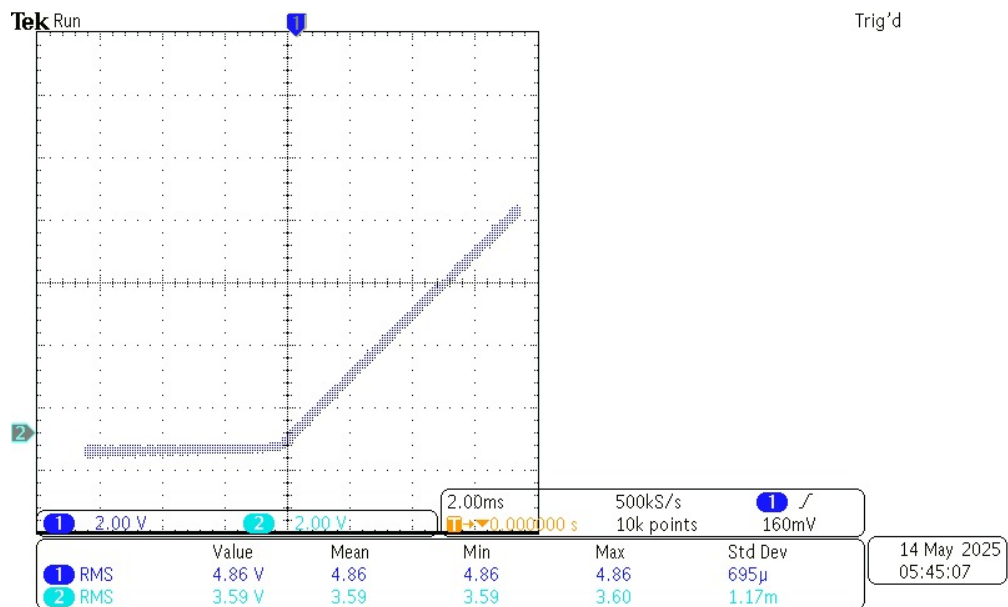


Why This Setup Ensures Safe Operation

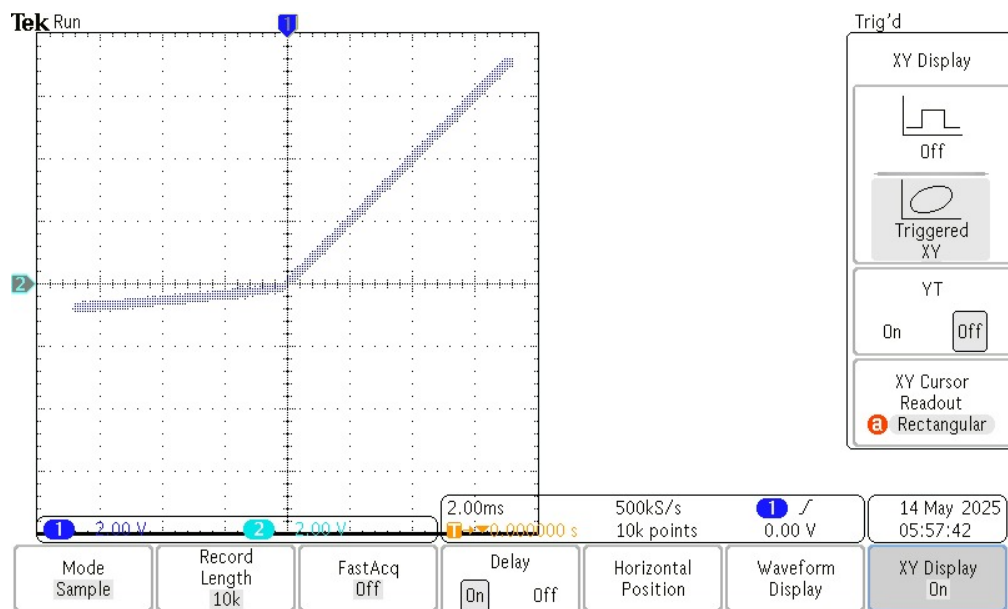
This method is particularly suitable and safe for diode testing, even in cases where the diode temperature increases, due to the following reasons:

- **Current Limitation:** The test circuit includes a series resistor, which limits the maximum current through the diode. This protects the diode from thermal damage, even if the forward voltage drops as the diode temperature rises.
- **Low Test Currents:** The selected measurement range of 0–10 mA lies well within the safe operating limits for both the AA138 germanium diode and the 1N4148 silicon diode.
- **Instant Visual Feedback:** The use of an oscilloscope in X-Y mode provides immediate visual observation of the diode's behavior. Any irregularities, such as thermal runaway or excessive current, can be promptly identified and the circuit can be adjusted or shut down as needed.

5 First Experiment



1N4148 (Silicon Diode)



AA138 (Germanium)

6 Lab Task 1: Forward Characteristics of Silicon and Germanium Diodes

6.1 Objective

The objective of this experiment is to measure and compare the forward current-voltage (I-V) characteristics of a silicon diode (1N4148) and a germanium diode (AA138). These characteristics were obtained using both circuit simulation and an oscilloscope in X-Y mode. The comparison provides insight into the conduction properties and dynamic behavior of both diode types.

Experimental Setup

A sinusoidal voltage source with an amplitude of 5 V and frequency of 50 Hz was applied in series with a 1 k Ω resistor and the diode under test. The circuit was constructed and simulated in LTspice, and the same setup was used in the lab for oscilloscope measurements.

- * The diode current, I_D , was inferred by measuring the voltage drop across the series resistor.
- * The diode voltage, V_D , was measured directly across the diode.
- * The oscilloscope operated in X-Y mode with V_D on the X-axis and I_D (calculated from resistor voltage) on the Y-axis.

Observations

The forward characteristics plotted from simulation and oscilloscope data show the expected exponential relationship between voltage and current for both diodes.

- * **Silicon diode (1N4148):** The conduction starts at approximately 0.6 V. As the forward voltage increases beyond this point, the current rises sharply, indicating a low dynamic resistance.
- * **Germanium diode (AA138):** The conduction begins earlier, around 0.2–0.3 V. However, the increase in current is more gradual, and the diode exhibits a higher forward resistance.

These observations are consistent with the known properties of silicon and germanium diodes: germanium has a lower threshold voltage, but silicon provides faster switching and better thermal stability.

6.2 Evaluation 1: Differential Resistance at $I_D = 2 \text{ mA}$

Method

The differential (or dynamic) resistance, r_d , at a specific current is defined as the local slope of the forward I-V curve:

$$r_d = \left. \frac{dV}{dI} \right|_{I=2 \text{ mA}}$$

Graphically, this was estimated by selecting two nearby points on the I-V curve around $I = 2 \text{ mA}$ and calculating the ratio $\Delta V / \Delta I$.

Results

* Silicon diode (1N4148):

- At $I_D = 2 \text{ mA}$, the corresponding forward voltage is approximately 0.68 V .
- A local voltage change of about 40 mV results in a current change of 2 mA .
- Hence, the differential resistance is estimated as:

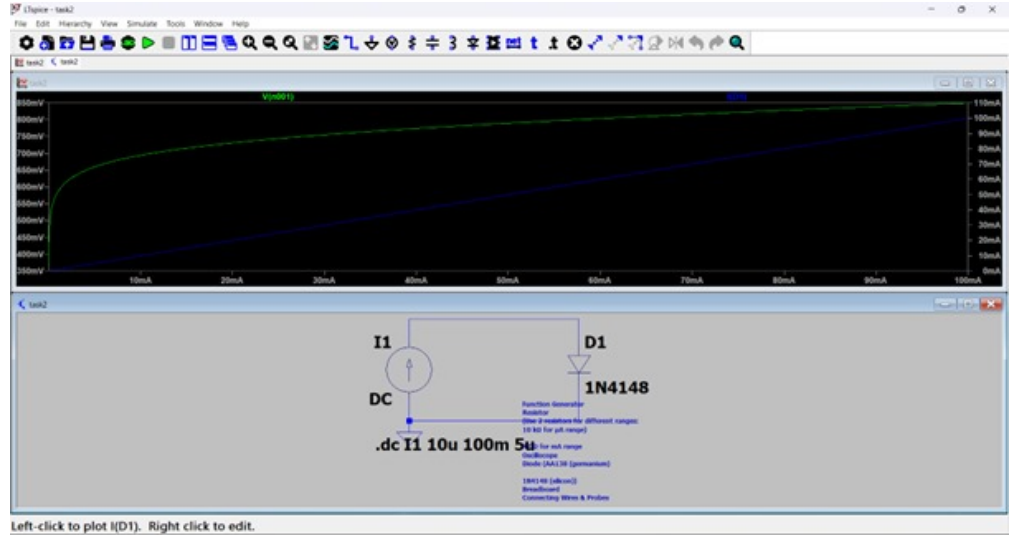
$$r_d^{(\text{Si})} \approx \frac{0.04 \text{ V}}{2 \text{ mA}} = 20 \Omega$$

* Germanium diode (AA138):

- At $I_D = 2 \text{ mA}$, the forward voltage is approximately 0.28 V .
- A local voltage change of about 80 mV results in a current change of 2 mA .
- Therefore, the differential resistance is estimated as:

$$r_d^{(\text{Ge})} \approx \frac{0.08 \text{ V}}{2 \text{ mA}} = 40 \Omega$$

7 Task 2 Preparation



Preparation 2: Diode Equivalent Model and Characteristic Derivations

1. Theoretical Background

The forward voltage of a real diode can be modeled using an equivalent circuit that combines the Shockley equation for the ideal diode and the voltage drop across the internal (bulk) resistance R_B . This leads to the expression:

$$U_F = U_{FD} + I_F \cdot R_B$$

Where:

- U_F is the total forward voltage across the diode,
- U_{FD} is the ideal diode forward voltage (without series resistance),
- I_F is the forward current,
- R_B is the bulk (series) resistance of the diode.

According to the Shockley equation for an ideal diode:

$$I_F = I_S \left(e^{\frac{U_{FD}}{nU_T}} - 1 \right)$$

Where:

- I_S is the reverse saturation current,
- n is the emission coefficient (typically between 1 and 2),
- U_T is the thermal voltage, given by $U_T = \frac{kT}{q} \approx 25.85 \text{ mV}$ at room temperature.

2. Deriving the Forward Voltage with Bulk Resistance

Substitute U_{FD} from the Shockley equation into the expression for U_F :

$$U_F = nU_T \ln \left(\frac{I_F}{I_S} + 1 \right) + I_F \cdot R_B$$

This is the complete expression for the forward voltage across a real diode.

3. Logarithmic Representation

To extract the diode parameters experimentally, the logarithmic form of the Shockley equation is used (valid when $I_F \gg I_S$):

$$\ln(I_F) \approx \frac{U_{FD}}{nU_T} + \ln(I_S)$$

This linear relationship in a semilogarithmic plot ($\ln(I_F)$ vs. U_{FD}) allows determination of:

- **Slope:** $\frac{1}{nU_T} \Rightarrow n = \frac{1}{\text{slope} \cdot U_T}$
- **Intercept:** $\ln(I_S) \Rightarrow I_S = e^{\text{intercept}}$

From the curvature at higher currents, R_B can be extracted based on the deviation from linearity due to the series resistance.

4. Theoretical Differential Resistance r_d

The differential (AC) resistance of the diode is the derivative of the forward voltage with respect to current:

$$r_d = \frac{dU_F}{dI_F}$$

Using the full expression:

$$U_F = nU_T \ln \left(\frac{I_F}{I_S} + 1 \right) + I_F R_B$$

Differentiate with respect to I_F :

$$r_d = \frac{nU_T}{I_F + I_S} + R_B \approx \frac{nU_T}{I_F} + R_B \quad (\text{if } I_F \gg I_S)$$

5. Summary of Extractable Parameters

- **Reverse saturation current** I_S : from the y-intercept of the $\ln(I_F)$ vs. U_F plot.
- **Emission coefficient** n : from the slope of the linear region of the semilog plot.
- **Bulk resistance** R_B : from the curvature in the high-current region.
- **Theoretical differential resistance** :

$$r_d = \frac{nU_T}{I_F} + R_B$$

Second Experiment

Measurement of Forward Characteristic of Silicon Diode (1N4148)

7.1 Lab Task 2

To measure the forward characteristic $I_D = f(U_D)$ of the silicon diode 1N4148 using two different resistors ($47\ \Omega$ and $10\ \text{k}\Omega$), across a current range of $10\ \mu\text{A}$ to $100\ \text{mA}$.

Equipment

- * Silicon diode: 1N4148
- * Resistors: $47\ \Omega$, $10\ \text{k}\Omega$
- * Signal generator
- * Oscilloscope (XY mode)
- * Breadboard and connecting wires

Measurement Setup

The diode is connected in series with a resistor and driven by a sinusoidal voltage source. The oscilloscope is configured in XY mode:

- * Channel 1 (X-axis): Measures voltage across the diode (U_D)
- * Channel 2 (Y-axis): Measures voltage across the resistor (U_R), to calculate current via $I_D = \frac{U_R}{R}$

Procedure

1. Configure the circuit for forward bias using the $10\text{ k}\Omega$ resistor.
2. Capture the diode's I-V curve using the oscilloscope in XY mode.
3. Repeat the measurement using the $47\text{ }\Omega$ resistor for higher current range.
4. Estimate and record voltage and current values from the characteristic curve.

Oscilloscope Readings

With $10\text{ k}\Omega$ Resistor

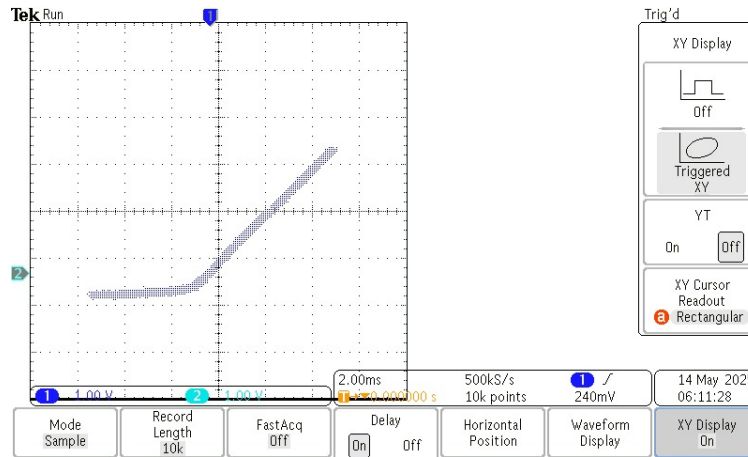


Figure 1: Forward characteristic of 1N4148 diode using $10\text{ k}\Omega$ resistor

With 47 Ω Resistor

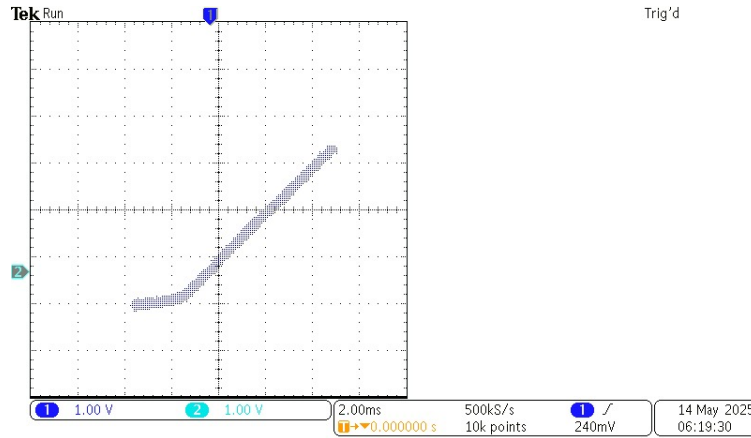


Figure 2: Forward characteristic of 1N4148 diode using 47 Ω resistor

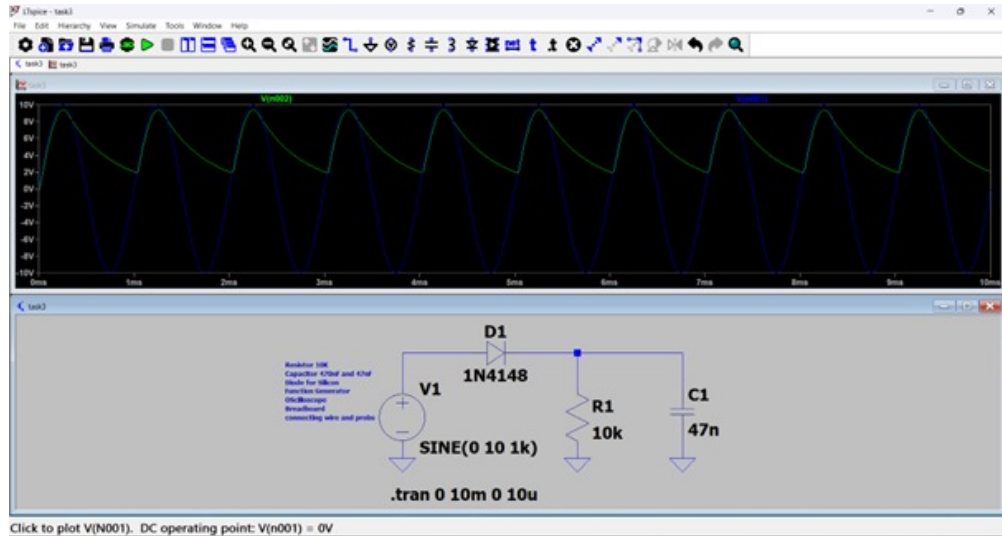
7.2 Results

Table 1: Estimated Forward Voltage and Current Values

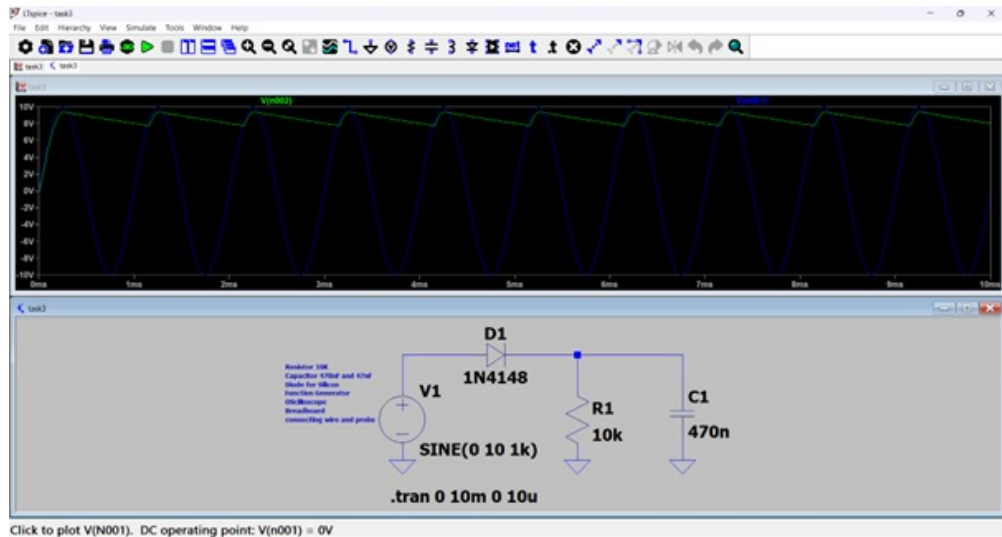
U_D (V)	I_D (10 k Ω) [mA]	I_D (47 Ω) [mA]
0.45	0.01	0.10
0.50	0.02	0.30
0.55	0.05	0.60
0.60	0.10	1.00
0.63	0.20	2.50
0.67	0.50	5.00
0.70	1.00	10.00
0.74	2.00	20.00
0.78	5.00	50.00
0.83	7.50	75.00
0.88	10.00	100.00

8 Task 3 Preparation

Half-wave rectifier circuit to investigate the ripple factor



Capacitor47nF



Capacitor470nF

Definition of Ripple Factor

The ripple factor r is a measure of the residual AC component (ripple) present in the rectifier output, relative to its average (DC) value. It indicates how effectively the rectifier smooths the signal. By increasing the capacitor, we obtain a more filtered DC signal.

Ripple Factor Formula

$$r = \frac{U_{AC,RMS}}{U_{DC}}$$

Where:

- $U_{AC,RMS}$ = RMS value of the **AC (ripple) component only**
- U_{DC} = **Mean value** of the rectified signal

Alternatively, using total (measured) RMS:

$$r = \sqrt{\left(\frac{U_{RMS}}{U_{DC}}\right)^2 - 1}$$

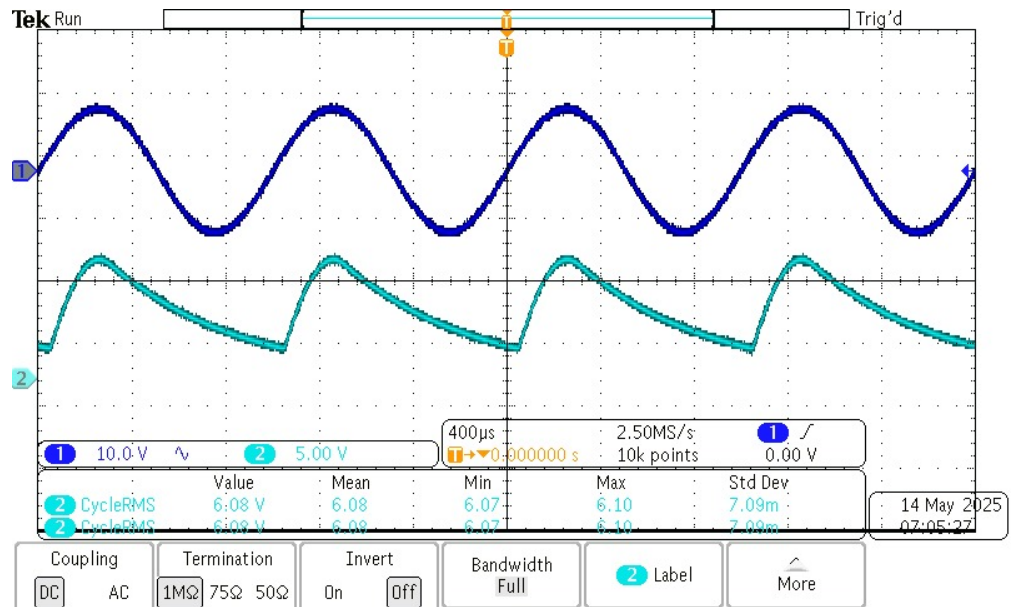
Where:

- U_{RMS} = Total RMS value of the output voltage U_2
- U_{DC} = Mean (average) value of the output voltage

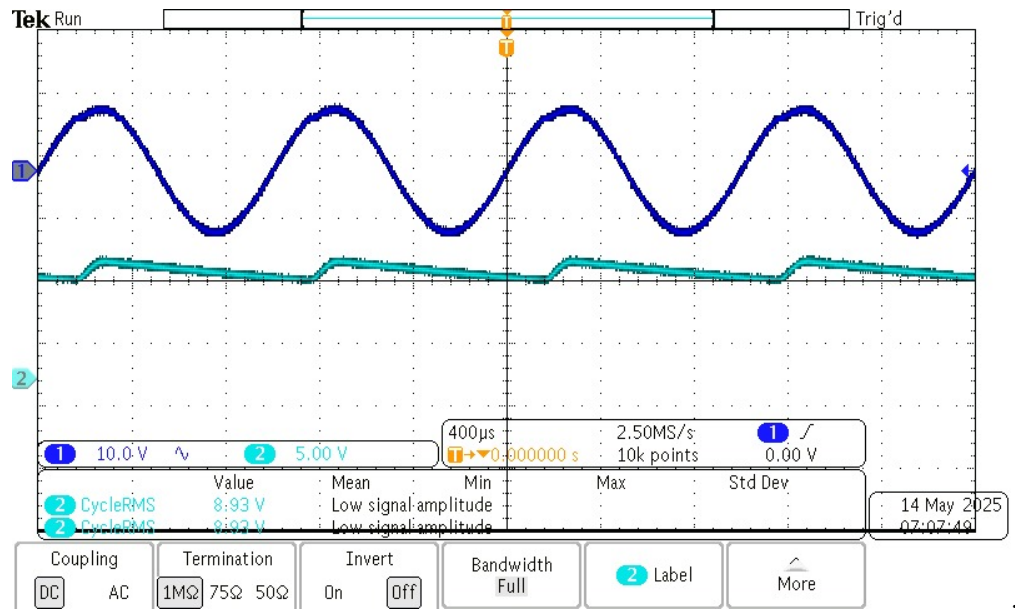
This equation comes from the fact that:

$$U_{RMS}^2 = U_{DC}^2 + U_{AC,RMS}^2$$

9 Third Experiment



Capacitor 47nF



Capacitor 470nF

9.1 Lab task 3

In this task, we built a half-wave rectifier circuit with two different capacitors: one with 47nF and another with 470nF. We measured the output voltage in both cases using an oscilloscope. Then we did the following:

Found the RMS voltage and the DC voltage from the output.

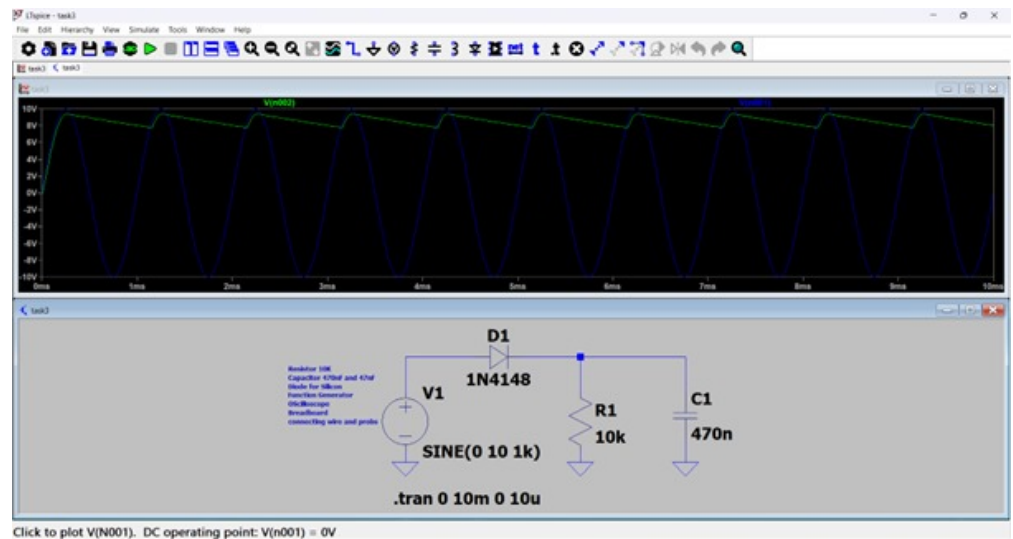
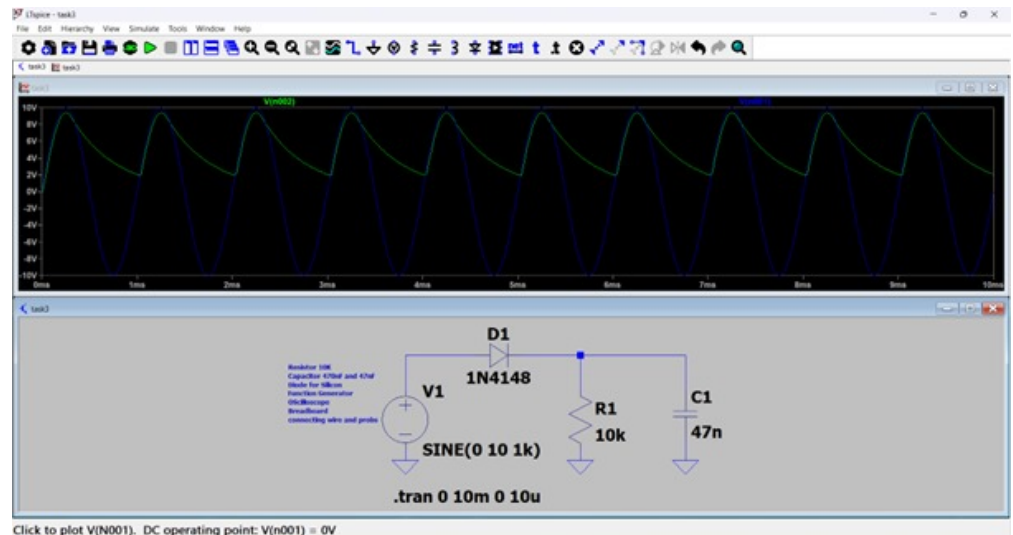
Used these values to calculate the ripple factor using two different formulas.

Compared the results to see how the capacitor value affects the ripple.

9.2 Evaluation

- Measured RMS and DC voltages for both capacitors
- Calculated ripple factors using the formulas above
- Comparison of the results
- Explanation of how increasing the capacitance affects the ripple
- Comments on signal smoothness
- Any observed anomalies or measurement errors

10 Task 4 Preparation



Use of Differential Probe

A differential probe is used to safely measure the voltage between two arbitrary points in a circuit, especially when neither of them is referenced to ground. It offers the following advantages:

- It allows measurement between two points without tying either to the oscilloscope's ground.
- Both ends of the measurement are electrically isolated from the oscilloscope, preventing ground loops or short circuits.

- It ensures an accurate view of the true differential signal without disturbing the circuit operation.

Why Not Use a Passive Probe?

Passive probes are referenced to the oscilloscope ground, which is typically connected to earth ground. Using a passive probe in a floating or differential configuration can cause:

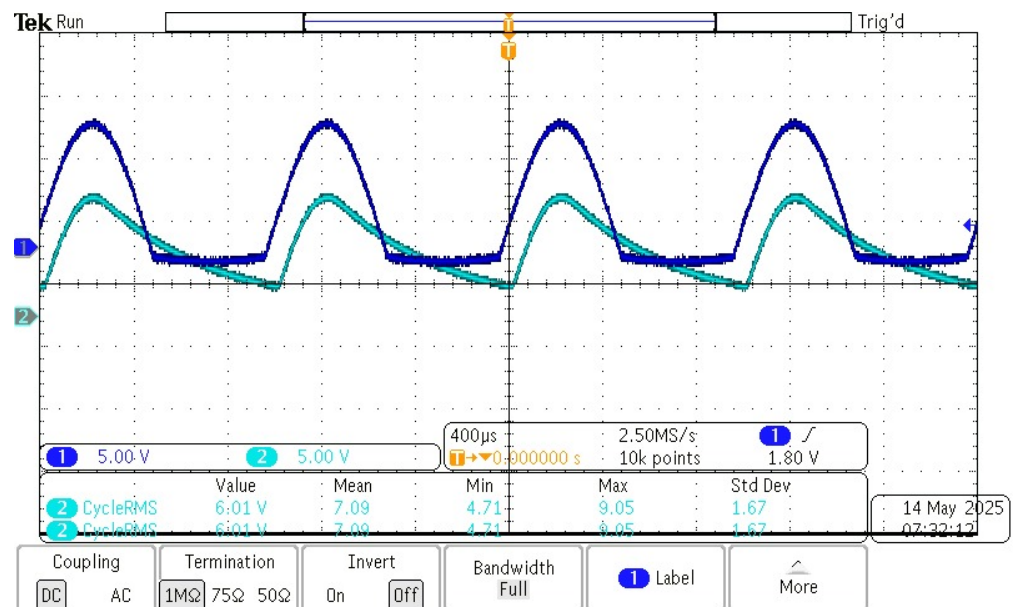
- **Short circuits or damage** to the circuit or equipment if one of the points being measured is not at ground potential.
- **Distortion of the measured signal** due to unintended grounding of the circuit.
- **Safety risks** if high voltages or non-isolated power supplies are involved.

For these reasons, a differential probe is essential for accurate and safe measurement in

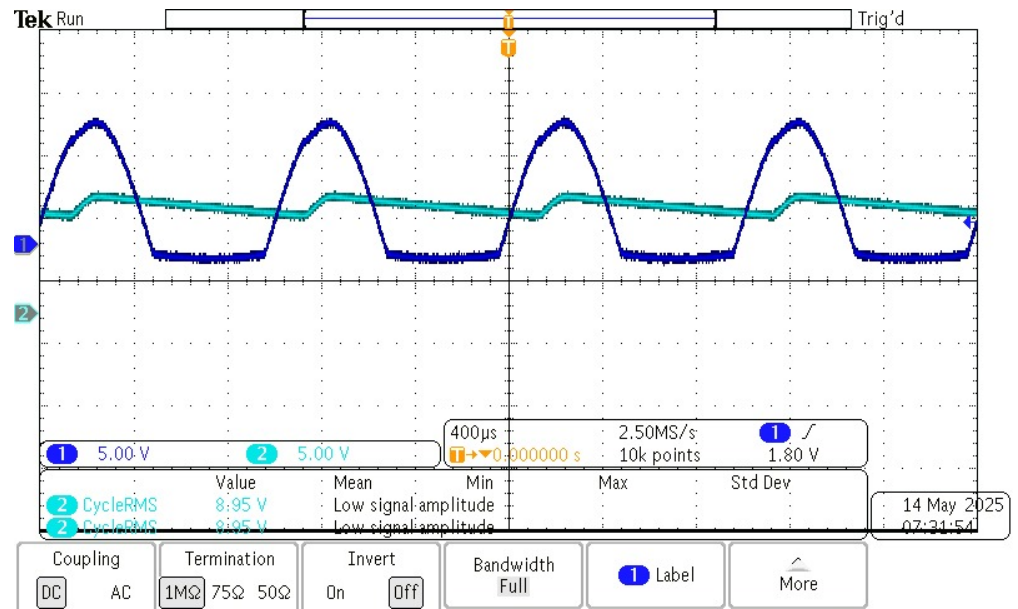
differential configurations.

11 Fourth Experiment

Full-wave rectifier circuit



Capacitor47nF



Capacitor 470nF

11.1 Lab task 4

In this task, we used a full-wave rectifier circuit with two different capacitors: one with 47nF and another with 470nF.

We did the following steps:

Connected the circuit and measured the output voltage with an oscilloscope.

Took the RMS value and DC value of the output.

Calculated the ripple factor using both formulas.

Compared how the ripple changed when we used a bigger capacitor.

This helped us understand how full-wave rectifiers and capacitors work together to make the output smoother.

11.2 Evalutaion

We saw that the ripple (the leftover AC part in the output) became smaller when we used a larger capacitor (470nF). This means the output got smoother and closer to a pure DC voltage.

Also, compared to the half-wave rectifier from Lab Task 3, the full-wave rectifier in this task gave a much more stable output with less ripple, even with the same capacitor values.

So, we learned that:

Full-wave rectifiers work better for reducing ripple.

Bigger capacitors help smooth the output even more.

12 Conclusion

In this lab, we explored the behavior and characteristics of semiconductor diodes through both practical experiments and simulations. By comparing a germanium diode (AA138) with a silicon diode (1N4148), we observed their forward I-V curves. As expected, the germanium diode started conducting at a lower voltage (0.2–0.3V), while the silicon diode required a higher threshold (0.6–0.7V), matching their typical behavior.

We also verified the Shockley diode equation using semilog plots and calculated important parameters like the ideality factor, saturation current, and series resistance. These values helped us better understand how real diodes differ from ideal ones, especially when it comes to internal resistance and overall performance.

During the rectifier part of the lab, we built and tested half-wave and full-wave rectifier circuits. Adding smoothing capacitors significantly reduced the ripple voltage, showing how effective they are in converting AC to a more stable DC output. The full-wave rectifier proved to be more efficient, with less ripple and better overall performance.

Overall, this lab helped us connect theoretical concepts about semiconductors with real circuit behavior, giving us a clearer understanding of how diodes are used in practical electronics.

13 Reference

Electronic Circuits: Handbook for Design and Application“, Ulrich Tietze / Christoph Schenk / Eberhard Gamm, Springer Vieweg
Electronic Devices and Circuit Theory“, Robert Boylestad / Louis Nashelsky, Pearson Education Limited